

Alison S.

Boston, MA 02115 | (+1) 781 288 9645 | alison.cs.dev@gmail.com | [linkedin.com/in/jiayongshi/](https://www.linkedin.com/in/jiayongshi/) | github.com/AlisonShi0410

EDUCATION

Master of Science in Computer Engineering, Tufts University, Medford, MA **Sep. 2023 – May. 2025**

- Core courses: Machine Learning, Deep Learning, Computer Vision, Information Theory, Computer Architecture, Communication Network, Programming, AI, Data visualization

Bachelor of Science in Computer Science, Beijing University of Agriculture, BJ, CN **Sep. 2019 – Jul. 2023**

SKILLS

- Languages & Frameworks: **Python, Java, C++, SQL, Bash, JavaScript, HTML/CSS**
- Software & Development: **Spring, RESTful APIs, MVC Architecture, Git, CI/CD, Docker, Linux, Tomcat Server**
- Machine Learning & Data: **TensorFlow, Scikit-learn, BERT, Pandas, NumPy, Spark, MySQL, ETL, SVM, KNN, Tableau, Pivot Tables**; Cloud & Infrastructure: **AWS (EC2), Azure, Shell Scripting, Architecture**
- Systems & Tools: **Performance Benchmarking, Sniper Simulator, Embedded Systems, CAD Tools, Testing**
- Microsoft Certified: Azure AI Fundamentals 2025**

TECHNICAL PROJECT

Data Engineer | Loan Risk Analysis Project, Reliance Capital Inc **Jun. 2025 – Present**

- Used **SQL** to analyze borrower behavior and uncover default risk patterns, supporting credit decision-making.
- Queried, cleaned, and joined over 100K rows across application, credit score, and repayment data using **MySQL**; addressed missing values, format inconsistencies, and duplicate IDs.
- Leveraged **Tableau** to create user dashboards that augment and enrich **analytical insights**.
- Built a reusable SQL report** for weekly risk tracking, reducing manual workload by 70% and informing underwriting threshold refinements for low-risk applicants.

ML Engineer | Sentiment Classification System Using TF-IDF and BERT

Dec 2024

Python, Scikit-learn, BERT, Logistic Regression

- Built a **multi-source sentiment classification** pipeline using **100K+ labeled reviews** from IMDb, Amazon, and Yelp; improved F1-score by 12% over baseline.
- Combined traditional **NLP (Bag-of-Words, TF-IDF)** with **BERT** embeddings to capture both lexical and contextual semantics, enabling robust performance across domains.
- Participated in daily stand-ups, raised implementation concerns, and integrated peer feedback during code reviews.
- Applied unit testing for custom tokenizer modules and maintained modular code structure for reusability and readability.

Software Engineer | Full-Stack E-commerce Website

Sep. 2022 – Jun 2023

Java, JSP, Spring, Hibernate, MySQL, Tomcat, HTML/CSS

- Built and deployed a full-stack e-commerce system using **SSH with MVC architecture**, completed the full development **lifecycle**, aligning with **software development life cycle (SDLC)** best practices.
- Designed interactive frontend components with **JSP and HTML/CSS**, and implemented session handling logic.
- Redesigned **MySQL** schema and queries to support high-throughput transactions with secure session control.
- Created test plans and performed validation checks aligned with **QA** standards, improving product stability.

WORK EXPERIENCE

Software Testing Intern , RoboRock, BJ, CN

Sep. 2022 - Feb.2023

- Automated cross-device firmware deployment via **Shell scripting**, reducing update time per device by 30%.
- Collaborate with team members**, designed and executed functional **unit testing workflows** across embedded hardware platforms. Documented test failures with clear reproduction steps, collaborated with developers on triage, and verified **bug fixes**.
- Diagnosed root causes of system anomalies through **structured log analysis**, improving issue triage and bug resolution.

Algorithm Engineer Intern , JINHER NETWORK, BJ, CN

Jun. 2022 – Sep. 2022

- Built custom **ETL pipelines** to preprocess and manage 10K+ labeled images across multiple categories, enabling high-quality input for DNN training workflows.
- Accelerated **TensorFlow model training on AWS EC2 with GPU optimization**, achieving a 40% reduction in convergence time through parallelization and batch tuning.
- Adapted and modularized **open-source deep learning architectures** for production pipelines, improving inference speed and deployment efficiency.